

Military Institute of Science and Technology (MIST)

Computer Science and Engineering (CSE)

CSE 301 Database Management System

Class Test# 3 Batch: CSI-24 Date: 03 Sep 2026

[CO4: Be familiar with relational database theory and be able to write relational algebra expressions for queries. (PO2 (H): Identify, formulate, research literature, and analyze complex Engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.)]

Full Marks: 20

Time: 40 Minutes

Question 1: 3NF vs. BCNF -- University Course Assignment

Consider the relation used by a university to record course sections:

Assign-Schema = (StudentID, CourseID, InstructorID, Room)

Business rules: A student can take many courses. An instructor teaches only one fixed course, always in one fixed room. Following Functional dependencies hold on the Assign-Schema:

$F = \{$ (StudentID, CourseID) $\rightarrow$ InstructorID,
InstructorID $\rightarrow$ CourseID,
InstructorID $\rightarrow$ Room $\}$

(a) Find all Candidate Keys of the Assign-Schema.

(b) Check each FD against the Superkey rule. State whether the Assign-Schema is in 3NF and whether it is in BCNF, justifying with the formal definitions.

(c) Consider the following instance of Assign-Schema:

StudentID	CourseID	InstructorID	Room
S01	CSE301	I01	Room101
S02	CSE301	I01	Room101
S03	CSE301	I01	Room101
S04	CSE302	I02	Room102

Explain what information is unnecessarily repeated here and describe one resulting update anomaly. Then decompose the Assign-Schema into BCNF relations, stating each schema and its key(s).

(d) Is your decomposition dependency-preserving for (StudentID, CourseID) $\rightarrow$ InstructorID? Justify using closure, and briefly explain what this implies for choosing BCNF vs. 3NF here.

(e) Compare between BCNF and 3NF. Discuss how to overcome the limitations of these two forms of normalization.

[Hints and Guidelines: Answers should connect each part's reasoning (keys $\rightarrow$ normal form $\rightarrow$ redundancy $\rightarrow$ decomposition $\rightarrow$ dependency preservation), not just state definitions]

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Question 2: Write a SQL for creating a Trigger called 'overdraft-trigger' for the following situation: 'suppose that, instead of allowing negative account balances, the bank deals with overdrafts by setting the account balance to zero, and creating a loan in the amount of the overdraft. The bank gives this loan a loan number identical to the account number of the overdrawn account. For this example, the condition for executing the trigger is an update to the account relation that results in a negative balance value.'

05

HELLO!

CT=4

Military Institute of Science and Technology (MIST)

Computer Science and Engineering (CSE)

CSE 301 Database Management System

Class Test# 3 Group# B Batch: CSE-24 Date: 24 Sep 2026

[CO4: Be familiar with relational database theory and be able to write relational algebra expressions for queries. (PO2 (H): Identify, formulate, research literature, and analyze complex Engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.)]

Full Marks: 20

Time: 40 Minutes

Question 1:

1. Compare between Binary Trees and Adel'son-Vel'skii and Landis (AVL) Trees for indexing. A B-tree of order 3 contains 14 keys of records. The keys of the records are 1, 2, 5, 7, 8, 10, 12, 13, 14, 18, 19, 21, 22 and 25. Draw the tree (Hints: root has key of record 12 and two non-terminal nodes have key of records (5, 8) and (18, 22) respectively). Insert new key of record 16 and 4 one after another in the tree (Draw diagram only, no explanation is required). [04+03+04=11]

Question 2:

2. "Every view serializable schedule is also conflict serializable, but there are conflict serializable schedules that are not view serializable." — do you agree with this statement? Why? Explain with example. "Blind writes appear in any view-serializable schedule that is not conflict-serializable."- Justify this statement with an example.

[04+05=09]

Military Institute of Science and Technology (MIST)

Computer Science and Engineering (CSE)

CSE 301 Database Management System

Class Test# 2 Batch: CSE-24 Date: 18 Aug 2026

[C02: Illustrate the basics of SQL and construct queries using SQL. C04: Be familiar with relational database theory and be able to write relational algebra expressions for queries. (PO2 (H): Identify, formulate, research literature, and analyze complex Engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.))

Full Marks: 20

Time: 40 Minutes

1. A database schema, along with primary key and foreign key dependencies, can be depicted pictorially by schema diagrams. Figure 1 shows the schema diagram for a banking enterprise.

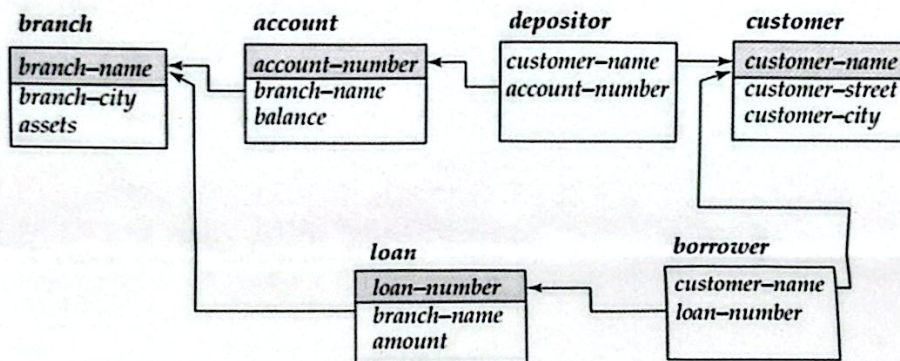


Figure 1: Schema Diagram for a Banking Enterprise

- a. Considering the above banking schema, briefly explain the “**Division (+) Operation**” in relational algebra. State the type of query for which the division operation is particularly useful. Using the banking database schema, write a “**relational algebra expression**” to find the names of all customers who have an account at every branch located in Brooklyn. Explain why the Division operation is appropriate for this query. **10**
- b. Considering the above banking schema, discuss the concept of “**Views**”. Write down the SQL statement for creating a “**View**”. Create a view consisting of branch names and the names of customers who have either an account or a loan at that branch. Assume that we want this view to be called “**all-customer**”. Using the view “**all-customer**”, find all customers of the “**Mirpur Branch**”. **10**

Military Institute of Science and Technology (MIST)

Computer Science and Engineering (CSE)

CSE 301 Database Management System

Mid Term Test# 1 Batch: CSE-24 Date: 23 July 2026

[C03: Analyze real life scenarios and suggest solutions based on the concepts and application of a Relational Database System with proper reasoning. (PO3 (H): Design/Development of Solutions, PO Description: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety as well as cultural, societal and environmental concerns.)]

Full Marks: 30

Time: 50 Minutes

The Scenario (read carefully)

City Care General Hospital wants a database for its core operations. Read the description below and design it — nothing here is purely decorative; every sentence contains something you are expected to model.

The hospital employs staff, each with a unique **employee ID**, a ^{complete} **full name** (first and last name), one or more **contact phone numbers**, and a **salary**. Every employee is either a **doctor**, for whom the hospital additionally records a **specialization** (e.g., Cardiology), or a **nurse**, for whom it records a **shift** (Morning/Evening/Night). The remaining employees are administrative staff, for whom the hospital records their designation (e.g., Accountant, Receptionist, Store Manager). A person cannot simultaneously be a doctor and a nurse, but every doctor, nurse, and administrative staff member is first and foremost an employee of the hospital, and shares all the general employee information described above. No employee is in two or more groups.

For emergency-contact purposes, the hospital also records each employee's **dependents** — such as a spouse or child. A dependent has no existence of its own in the records; it is only ever known through the employee it belongs to, and the hospital only needs a dependent's name and relationship to the employee (e.g., spouse, son).

Some doctors are still in training and are supervised by a senior doctor. The hospital wants to record which senior doctor supervises which junior doctor(s), along with the **date supervision began**. A senior doctor may supervise several junior doctors, but each junior doctor has only one supervisor.

Patients are registered with a unique **patient ID**, **name**, and **date of birth**. The hospital does not store a patient's age directly — it should always be calculated from the date of birth on file.

When a doctor treats a patient, the hospital records the diagnosis and the date of that consultation. A patient is typically treated by more than one doctor, and a doctor treats many patients.

The hospital also tracks medicine prescriptions. When a doctor prescribes a medicine to a patient, the hospital must record the dosage, and the date the prescription was written. Note that the same doctor may prescribe the same medicine to the same patient again on a different date for a different reason — so knowing only "this doctor prescribes this medicine" or only "this patient takes this medicine" is not enough; the fact connects doctor, patient, and medicine together all at once.

Finally, some treatments are submitted to an insurance company for reimbursement. When this happens, the insurance company is not approving a doctor or a patient by themselves — it is approving one specific instance of a doctor having treated a patient. For each such approval, the hospital records a claim ID and the amount approved. Not every treatment is submitted for approval.

Task A — E-R Diagram

Draw a complete E-R diagram for the scenario above. Your diagram will be marked on whether it correctly and clearly shows (if applicable):

1. Strong and weak entity sets, with keys/partial keys identified.
2. Correct identification and notation of composite, multivalued, and derived attributes.
3. A generalization/specialization hierarchy with the correct constraint (disjoint/overlapping, total/partial).
4. A recursive relationship with both roles clearly labeled.
5. A ternary relationship with its descriptive attribute(s) attached correctly.
6. An aggregation, used where a relationship itself needs to participate in another relationship.

Task B — Relational Schema Diagram

Starting from your own E-R diagram, map it to a relational schema. Show every relation with its attributes, primary key underlined, and foreign keys marked. Pay particular attention to how you map the weak entity, the specialization, the recursive relationship, the ternary relationship, and the aggregation — these carry most of the marks.

Task C — Assumptions

List, in short bullet points, any assumption you made to resolve ambiguity in the scenario.

Military Institute of Science and Technology (MIST)

Computer Science and Engineering (CSE)

CSE 301 Database Management System

Mid Term Test# 1 Batch: CSE-24 Date: 29 July 2026 (Group#B)

[C03: Analyze real life scenarios and suggest solutions based on the concepts and application of a Relational Database System with proper reasoning. (PO3 (H): Design/Development of Solutions, PO Description: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety as well as cultural, societal and environmental concerns.)]

Full Marks: 30

Time: 50 Minutes

The Scenario (Read and do Brain Storming carefully: — 10 minutes)

Stitch Craft Apparel Ltd., a readymade garments manufacturer, wants a database for its core operations. Read the description below and design it — nothing here is purely decorative; every sentence contains something you are expected to model.

The company employs staff, each with a unique **employee ID**, a **full name** (first and last name), one or more **contact phone numbers**, a **date of joining**, and a **salary**. Every employee is either a **tailor**, for whom the company additionally records a **machine specialization** (e.g., Overlock, Flatlock), or a **cutting master**, for whom it records an **area of specialization** (e.g., Bulk Cutting, Pattern Cutting). The remaining employees are **administrative staff**, for whom the company records their **designation** (e.g., Accountant, Store Keeper). A person cannot simultaneously be a tailor and a cutting master, but every tailor, cutting master, and administrative staff member is first and foremost an employee, and shares all the general employee information above. No employee is in two or more groups. The company does not store how long someone has worked there directly — it should always be calculated from their date of joining.

Skill is passed down on the shop floor: an experienced tailor is often assigned to mentor a newly hired tailor during their first few months. The company wants to record *which tailor mentors which other tailor(s)*, along with the *date the mentorship began*. One tailor may mentor several others, but a given tailor is mentored by only one person at a time.

W // Each customer **order** has a unique **order ID** and an **order date**. An order is never for just one thing — it always lists one or more **line items**, and a **line item** has no **meaning** or **identity** outside the order it belongs to (it is only ever referred to as "item 2 of order 1057"). Each line item specifies a **size** and a **quantity**.

The company stocks **fabric**, each identified by a **fabric ID** and a **fabric name**. When a cutting master cuts a specific fabric for a specific order line item, the company records the **quantity cut** (in meters) and the **date** it was cut. The same cutting master may cut the same fabric for the same line item again on a different date for a different batch, so the fact ties the cutting master, the fabric, and the order line item together all at once.

Tailors then stitch the cut pieces against specific order line items, and for each such instance the company records the **quantity stitched** and the **date assigned**. Not every stitching assignment is checked, but some are audited by external **buyers** (the brands the garments are made for), each identified by a **buyer ID** and **name**. When this happens, the buyer is not auditing a tailor or an order item individually — it is reviewing one specific instance of a particular tailor having stitched a particular line item. For each such audit, the company records an **audit ID**, the **result** (Pass/Fail), and the **audit date**.

Task A — E-R Diagram: (18 marks, ~25 minutes)

Draw a complete E-R diagram for the scenario above. Your diagram will be marked on whether it correctly and clearly shows (if applicable):

1. *Strong and weak entity sets, with Primary keys/Partial keys identified.*
2. *Correct identification and notation of composite, multivalued, and derived attributes.*
3. *A generalization/specialization hierarchy with the correct constraint (disjoint/overlapping, total/partial) across all three employee types.*
4. *A recursive relationship with both roles clearly labeled.*
5. *A ternary relationship with its descriptive attribute(s) attached correctly.*
6. *An aggregation, used where a relationship itself needs to participate in another relationship.*

Task B — Relational Schema Diagram: (10 marks, ~12 minutes)

Starting from **your own** E-R diagram, map it to a relational schema. Show every relation with its attributes, primary key underlined, and foreign keys marked. Pay particular attention to how you map the weak entity, the specialization, the recursive relationship, the ternary relationship, and the aggregation — these carry most of the marks.

Task C — Assumptions: (2 marks, ~3 minutes)

List, in short bullet points, any assumption you made to resolve ambiguity in the scenario.

Military Institute of Science and Technology (MIST)

Computer Science and Engineering (CSE)

CSE 301 atabase Management System

Class Test# 1 Batch: CSE-24

[C01: Describe the basic concepts, applications, and design of database systems. (PO1 (H): Engineering Knowledge)]

Full Marks: 20

Time: 40 Minutes

1. A modern DBMS is designed as a layered and modular software system to achieve data independence, security, concurrency, and efficient query processing. Assume you are designing the architecture of a university database management system that serves students, faculty members, application developers, and database administrators.

Using a clearly labeled “Database System Structural” diagram, explain how the major functional components of a DBMS cooperate to execute a database request. Furthermore, evaluate how different classes of users access the database through appropriate user interfaces and describe the end-to-end processing of a database query from user submission to data retrieval. Your explanation should include:

- ✓ The roles and interactions of the Storage Manager, Query Processor, and Disk Storage, and explain how each component contributes to efficient database operation..
- ✓ Explain the functions of each internal component of the Storage Manager.
- ✓ The functions of the Query Processor and how it processes SQL queries.
- ✓ Discuss the purpose of the data files, data dictionary, and indices.
- ✓ Compare the interactions of different types of database users with the DBMS through their respective interfaces.
- ✓ Demonstrate the complete data flow during the execution of a typical SQL query.

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